Department of Computer Science

Academic Year	**2026**
Semester	☐ Fall ☒ Spring ☐ Summer
Course Code - Name	CPS 240 – Object Oriented Programming, A&D
Instructor	Dr. Razi Iqbal
Assessment	Assignment 3
Deadline	April 6th, 2026

Assignment 3

The main purpose of this assignment is to test your knowledge of Java GUI.

Instructions:

- You are required to submit your source (.java) files on Blackboard.
- Students having exactly similar code will get a straight 0.
- Comment on your code appropriately. Failing to do so would result in losing marks.
- The deadline for submission of this assessment is **April 6th, 2026**.
- A late penalty of 10% per day late will be applied, in the absence of a medical note, to a maximum of 2 days late. After 2 days, the assignment will not be accepted.

Question

Create a JavaFX GUI that reads student data from a CSV file (students_data.csv) and displays it using a **`ListView`** with custom cells. Users should be able to select and delete records from the list.

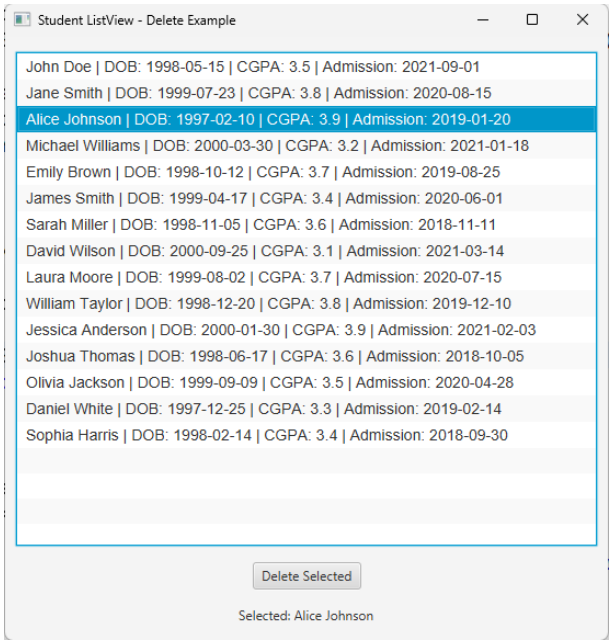
Please note the following:

- You are most welcome to read the CSV file any way you like, e.g., **`BufferedReader`**.
- Make sure to create a **`Student`** class with the following attributes:
  - **`firstName`** (`String`)
  - **`lastName`** (`String`)
  - **`dob`** (`String`)
  - **`cgpa`** (`double`)
  - **`admissionDate`** (`String`)

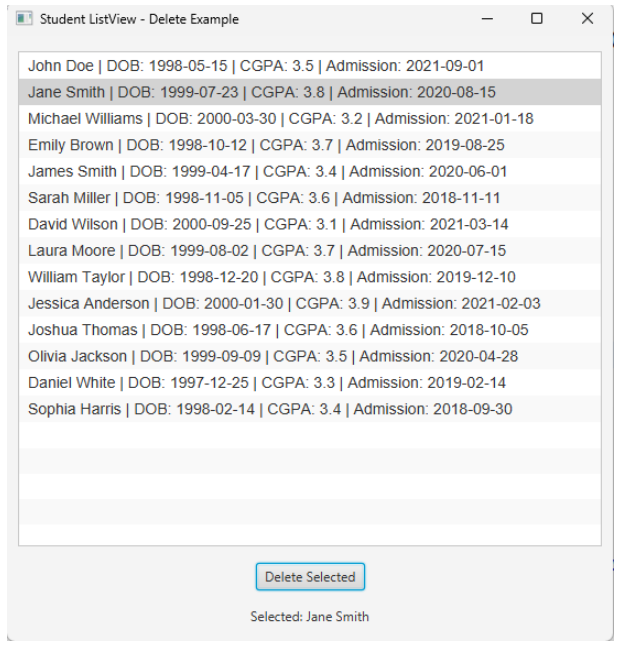
- Make sure the **`Student`** class has all the appropriate getters and setters.
- Read data from the csv file into **`Student`** objects. It is highly recommended to create a separate function for reading from the CSV file.
- You are most welcome to use any Layout to design your GUI but keep your design close to the one shown in the screenshots below.
- Use **`ListView<Student>`** to create a ListView of student objects.
- Make sure to show all student details in each row of the ListView as below. Please note the pipeline ("|") symbol and text tags like "DOB:", "CGPA:" and "Admission:".

```
John Doe | DOB: 2000-01-01 | CGPA: 3.5 | Admission: 2022-09-01
```

- You must use `setCellFactory(...)` to set font to size 14 and apply text color of your choosing to the rows of the ListView.
- ListView must handle single and multiple selection.
- When user clicks on a row of the ListView, the name of the student (first and last name) should appear on the Label as shown in the screenshot.
- When user clicks on a row and clicks "Delete Selected", that student row should be deleted from the ListView. (make sure your program allows multiple student deletion as well).
- It is a good idea to handle null values as well, e.g., no student in the ListView.
- Below are the screenshots of the expected outputs of before and after deletion.



Before Deletion



After Deletion

Rubric

- Reading CSV file (1 mark)
- Loading student data to ListView (1 mark)
- Deleting students (single, multiple) (2 marks)
- Cell Styling (1 mark)